Palaeomagnetic and Rock Magnetic Studies on Some Bulgarian Plio-Palaeocene Basalts

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Summary – Normal and reversed remanent magnetization has been found in Plio-Pleistocene basalts from Bulgaria. Ore microscope observations show that the magnetic mineral is a homogeneous 'titanomagnetite': electron microprobe studies show that it contains a high proportion of Ti. Comparison of thermomagnetic measurements with the Curie points expected for stoichiometric titanomagnetites with the analysed chemical composition suggest that the titanomagnetites must be slightly oxidized, i.e. be slightly cation deficient.

1. Introduction

These basalts were collected when one of us (KMC) visited Bulgaria as part of the programme of co-operation between the Royal Society of London and the Bulgarian Academy of Science. A sketch map of the sampling area is presented in Fig. 1.

In this paper we discuss their main magnetic characteristics describing measurements of natural remanent magnetization (N.R.M.), alternating field (A.F.), demagnetization, thermomagnetic curves, susceptibility and chemical composition of the titanomagnetites contained.

One interesting feature of these basalts is that they contain homogeneous titanomagnetites as their principal magnetic mineral.

2. N.R.M. and A.F. demagnetization

The direction and intensity of the natural remanent magnetization have been measured with a spinner magnetometer (DE SA and MOLYNEUX [4][4])). Each specimen was demagnetized in an alternating magnetic field (CREER [10]) to isolate the primary remanence and so to obtain the direction of the ancient geomagnetic field.

2.1. N.R.M. results

All samples from site 1 (the quarry near Chatal Hill) have a normal magnetization and the directions are well grouped as illustrated in Fig. 2 where they are represented

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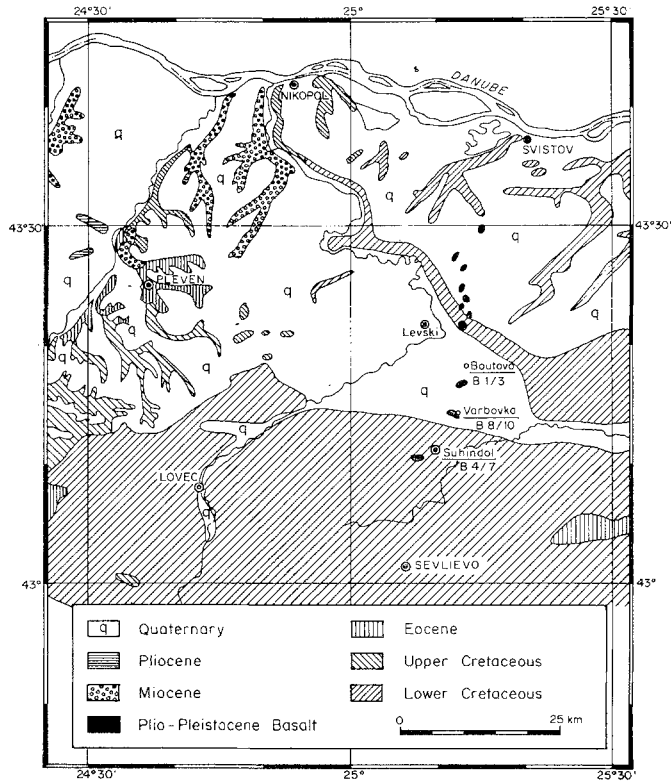


Figure 1
Geological sketch of the sampling area. Three samples were collected from each of sites 1, 2 and 3

as circles. On the other hand, samples from site 2 (Suhindol) are reversely magnetized and the directions, except for some specimens of B6 are more scattered. In Fig. 2 these directions are represented by squares.

In Table 1, we summarize these N.R.M. data and note that the mean intensity of the magnetization at site 2 is about three times that at site 1. Sample B7 with a high intensity comes from a weathered level near the surface.

Table 2 gives the N.R.M. values for site 3, for which the samples consisted of non-oriented core samples. Therefore we cannot deduce the angle between the horizontal component of N.R.M. and true north. The inclinations are negative and we infer from this that all these samples are reversely magnetized.

2.2. A.F. demagnetization results

One specimen from each sample from sites 1 and 2 was submitted to complete

Table 1

Bulgarian basalts, N.R.M.

Site	Locality	Sample No.	Number of specimens	D	I	α	δ	Mean Intensity 10^{-3} gauss	Mean + SD	Mean − SD
1	Quarry at Chatal	B1	6	10.9	67.6	10.2	12.2	2.71	3.37	2.17
1	Hill near	B2	4	26.1	62.4	16.0	14.0	2.84	3.61	2.23
	Boutovo village	B3	6	342.0	44.4	7.6	9.1	2.88	3.10	2.67
Site 1	mean	$N_d =$	16	359.4	59.0	8.5	18.2	2.80	3.34	2.36
2	Hill near	B4	6	151.7	−17.2	5.4	6.6	9.02	10.04	8.10
	Suhindol village	B6	6	18.7	5.5	18.5	21.7	4.14	6.33	2.71
2		B7	6	134.6	−19.6	3.8	4.5	25.94	29.20	2.30
Site 2	mean	$N_d =$	18	341.0	−16.0	13.6	30.0	9.90	24.85	3.94

D = Declination, I = Inclination, α = Radius of 95% of confidence about the mean, δ = Circular standard deviation, SD = Standard deviation, N_d = Number of data.
Sample and Site means are calculated from a geometrical distribution.

Table 2

Bulgarian core samples (non orientated), N.R.M.

Site	Locality	Sample No.	Number of specimens	I	α	δ	Mean Intensity 10^{-3} gauss	Mean + SD	Mean − SD
3		B8	6	−78.2	22.7	26.2	2.91	4.33	1.96
3	Hill north of Varbovka village	B9	6	−76.1	29.0	32.7	1.12	1.43	0.88
3		B10	6	−82.2	42.3	44.5	0.87	1.14	0.65

Legend see Table 1.
Note that the Declinations are unknown because the cores were unorientated in azimuth.

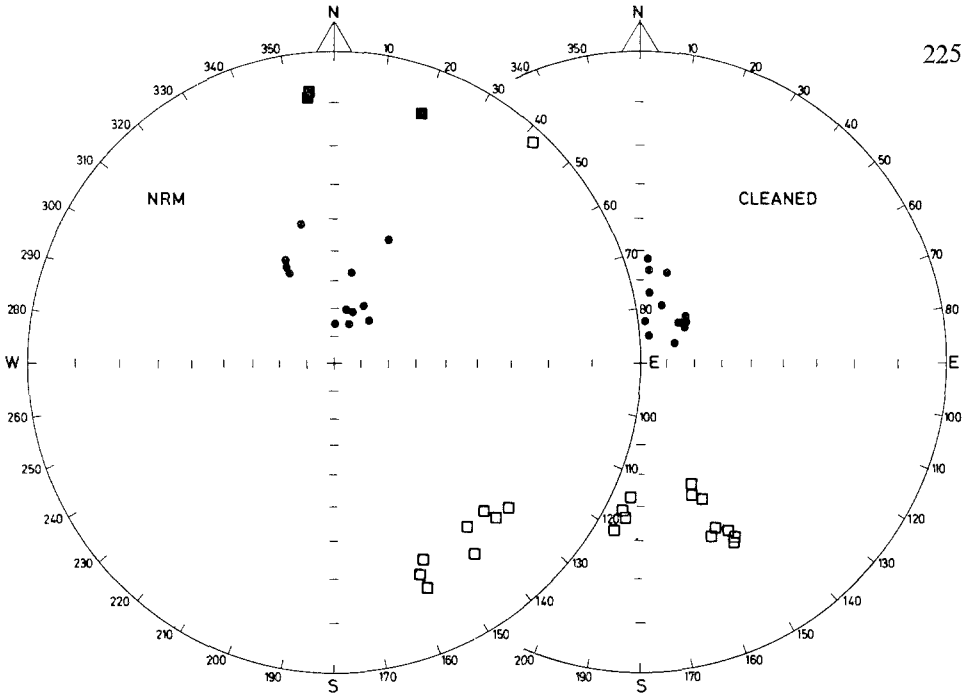


Figure 2

Remanent magnetization directions of the specimens frome site 1 (circles) and site 2 (squares) before and after partial demagnetization. The open symbols represent negative inclinations and the full ones positive inclinations. The directions are plotted on a stereographic projection (Wulff net)

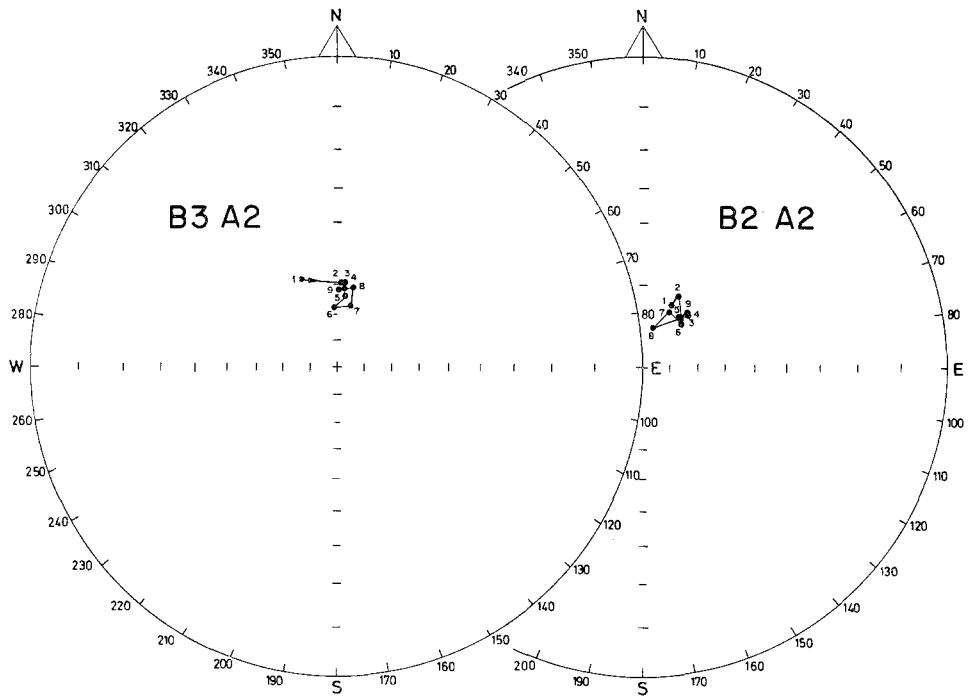


Figure 3

Examples of the response to alternating magnetic field demagnetization for site 1. The numbers represent the different steps as follows: — 1 = 0 Oe, 2 = 34 Oe, 3 = 68 Oe, 4 = 102 Oe, 5 = 135 Oe, 6 = 170 Oe, 7 = 212 Oe, 8 = 255 Oe, 9 = 340 Oe

stepwise alternating field cleaning. This procedure randomizes the effect of low coercivity components associated with the unwanted secondary magnetization.

(a) *Changes in direction of R.M.* – The normally magnetized samples from site 1 exhibit only small changes in direction of remanence during stepwise A.F. treatment, as shown in Fig. 3 for samples B3A2 and B2A2.

The reversely magnetized samples from site 2 exhibit larger changes in direction, as shown in Fig. 4 for samples B6A2 and B7A2, becoming more tightly grouped.

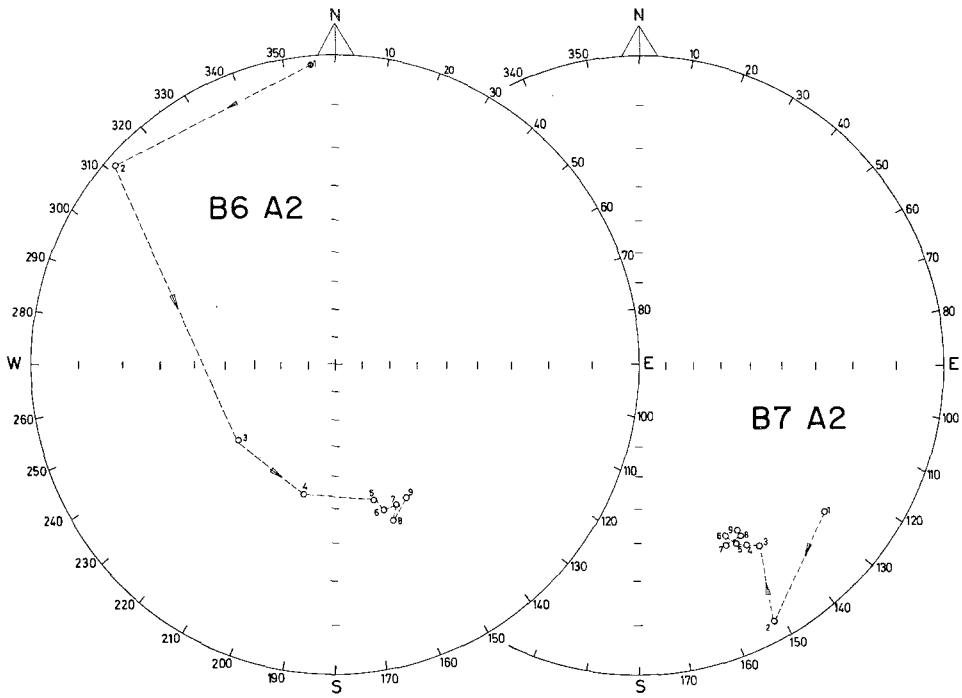


Figure 4
Examples of the response to alternating magnetic field demagnetization for site 2. The numbers represent the different steps as in Fig. 3

(b) *Demagnetization curves of the N.R.M. intensity* – The rapid decay of the R.M. in fields of less than 100 Oe indicates that these basalts have an important soft component of magnetization. The A.F. coercivity (ZIJDERVELD [23]) defined as the alternating field that eliminates just half of the magnetization, is in the range of 25 to 55 Oe (peak values). There is no significant difference between samples with normal and reversed polarity: it is just possible that the former may show a higher 'coercivity' at 20% of the original magnetization.

Table 3

Bulgarian basalts, R.M. after A.F. cleaning

Site	Sample No.	Number of Specimens	D	I	α	δ	Mean Intensity 10^{-3} gauss	Mean +SD	Mean −SD	North Pole Lat	North Pole Long	Polarity
1	B1	4	28.7	75.2	9.5	8.3	0.58	0.84	0.40	64.5	56.4 E	N
1	B2	4	37.8	68.1	2.2	1.9	0.62	0.84	0.45	63.6	85.2 E	N
1	B3	4	8.9	57.1	6.7	5.9	0.51	0.58	0.46	81.3	151.9 E	N
Site 1 mean N_s=		3	22.4	67.4	17.0	11.1	0.59	0.66	0.52	72.4	80.8 E	N
Site 1 mean N_d =		12	22.4	67.4	5.9	10.9	0.57	0.62	0.52	72.4	81.0 E	N
2	B4	4	186.5	−37.9	5.0	4.4	1.28	1.38	1.18	67.2	189.8 E	R
2	B6	4	155.0	−36.3	10.7	9.4	1.01	1.14	0.90	59.0	255.4 E	R
2	B7	4	154.0	−26.0	3.6	3.2	3.16	3.47	2.88	52.8	250.3 E	R
Site 2 mean N_s =		3	164.5	−34.3	25.5	15.6	1.60	2.92	0.88	62.5	237.7 E	R
Site 2 mean N_d =		12	164.5	−34.3	8.2	15.1	1.60	2.91	0.88	62.5	237.7 E	R

Legend see table 1.
N = Normal polarity, R = Reversed polarity

3. *Palaeomagnetic results*

All oriented specimens from sites 1 and 2 were subjected to A.F. cleaning in two steps (68 and 135 Oe), deduced from the pilot specimen analysis. The results, given in table 3 should be considered as preliminary because of the small number of hand samples. Nevertheless they provide some useful information. Each of the two sites exhibits a different geomagnetic polarity. Site 1 possesses normal remanent magnetiza-

Table 4

Bulgarian basalts, intensity susceptibility ratios

Site	Sample No.	Number of specimens	J_0 10^{-3} gauss	J 10^{-3} gauss	J/J_0	κ 10^{-3} gauss/Oe	J_0/κ
1	B1	6	2.71	0.58	0.22	3.1	0.9
1	B2	4	2.84	0.62	0.22	2.7	1.1
1	B3	6	2.88	0.51	0.18	2.2	1.3
Site 1 mean $N_s=$		3	2.80	0.59	0.21	2.6	1.1
Site 1 mean $N_d=$		16	2.80	0.57	0.20	2.6	1.1
2	B4	6	9.02	1.28	0.14	0.7	12.9
2	B6	6	4.14	1.01	0.24	0.7	5.9
2	B7	6	25.94	3.16	0.12	1.0	25.9
Site 2 mean $N_s=$		3	10.18	1.60	0.16	0.8	12.7
Site 2 mean $N_d=$		18	9.90	1.60	0.16	0.8	12.4
3	B8	6	2.91	–	–	1.8	1.6
3	B9	6	1.12	–	–	1.6	0.7
3	B10	6	0.87	–	–	1.7	0.5

J_0 = Intensity of NRM
J = Intensity of cleaned R.M.
κ = Volume susceptibility

tion whereas site 2 possesses reversed remanent magnetization. Our basalts are known to be Plio-Pleistocene, but no radiogenic age determinations have yet been made. The absence of an accurate age does not allow the correlation of these polarities with two specific geomagnetic epochs, several earth magnetic field reversals having been described for the upper Cenozoic (DALRYMPLE *et al.* [11]).

Site 1

The three samples from near Boutovo Village are grouped around an average direction of magnetization with $D=22.4°$, $I=+67.4°$ with a radius of the 95% circle of confidence about the mean (FISHER [13]) $\text{alpha}_{95}=17.5°$ assigning unit weight to each sample ($N=3$) and $\text{alpha}_{95}=5.9°$ if we assign unit weight to each disk cut from the samples ($N=12$).

Site 2

The directions of the three samples from near Suhindol Village are grouped around an average value with $D=164.5°$, $I=-34.5°$ and $\text{alpha}_{95}=25.5°$, ($N=3$) and $\text{alpha}_{95}=8.2°$ ($N=12$), assigning unit weight to samples and disks respectively. If we compare the two sites, the first yields the more coherent data. The larger within site scatter of sample directions at site 2 is probably related effects of weathering, e.g. note the high intensity magnetization of sample B7.

4. Natural magnetic properties

The properties investigated listed in Table 4 are the intensity of natural remanent magnetization (J_0), the low field susceptibility (κ) and the ratio of the natural remanent magnetization to the susceptibility, (similar to the Koenigsberger ratio with unit field).

The N.R.M. intensities vary from 0.87×10^{-3} to 25.94×10^{-3} Gauss. Except for the latter high value which was derived from a weathered sample, the J_0 data are very similar to other continental or submarine basalts (ADE-HALL [2], TARLING [19], OZIMA *et al.* [17].)

The susceptibilities vary between 0.7×10^{-3} to 3.1×10^{-3} Gauss/Oersted. These values are higher and consequently, the ratio N.R.M./κ is lower than for submarine basalts which have similar thermomagnetic properties to out basalts. It is interesting to note that our lowest ratio coincides with the core samples which are the deepest investigated.

5. Ore microscope observations

Twelve specimens were observed under oil by reflected light to detect the optical oxidation class (ADE-HALL *et al.* [1]). The titanomagnetites were homogeneous but two types of grain were noted. The most common, more or less regularly distributed, is characterized by small subhedral shape. The second type, having a large skeletal form, occurs occasionally in a polished section.

From these observations it is possible to distinguish two generations: the smaller having been formed later. Some large titanomagnetites have a grey core (WRIGHT [22], BABKINE *et al.* [4]). Ilmenite has been found as an isolated phase. One can therefore deduce that the titanomagnetite and the ilmenite are in equilibrium.

6. Determination of the chemical composition of the titanomagnetites

Five specimens were selected for investigation by the electron microprobe (Geoscan). The analysis made on different grains are given in Table 5. The determination of 'compositional' Curie points has been made supposing the grains to be stoichiometric, but impure titanomagnetites. The amount by which the presence of trace

Table 5

Bulgarian basalts, Electron microprobe analyses: site 1

Sample No.:		B1	B3	B6	B9	B10
Number of analyses:		3	3	3	4	1
Degree of oxidation:		1	1	1	1	1
	Fe	53.5	47.9	52.1	50.0	51.6
	Ti	10.3	13.9	11.6	13.3	14.4
	Mg	1.9	2.9	3.2	1.2	0.7
	Al	3.6	4.1	2.1	3.0	3.0
	Cr	0.3	0.4	0.4	2.4	0.1
	Mn	0.5	0.7	0.7	0.5	0.5
	f	1.06	1.06	1.09	1.05	1.05
	C	70.1	69.7	70.1	70.4	70.3
	R	0.190	0.285	0.218	0.251	0.276
	x	0.47	0.61	0.54	0.60	0.64
	$T_c(1)$	250°C	150°C	200°C	160°C	125°C
	$T_c(2)$	205°C	85°C	165°C	125°C	100°C

f: normalization factor used to make total weight % of cations = C.
C: total weight % of cations expected for unoxidized titanomagnetite containing the proportions of trace elements analysed.
R: ratio between weight % of Ti to the sum of the weights % of Fe, Mn and Cr.
x: molecular proportion of ulvöspinel in the pure titanomagnetite containing the above weight % of Ti.
$T_c(1)$: Curie point expected for the pure titanomagnetite with compositional parameter x.
$T_c(2)$: Estimated Curie point expected for the impure titanomagnetite analysed.

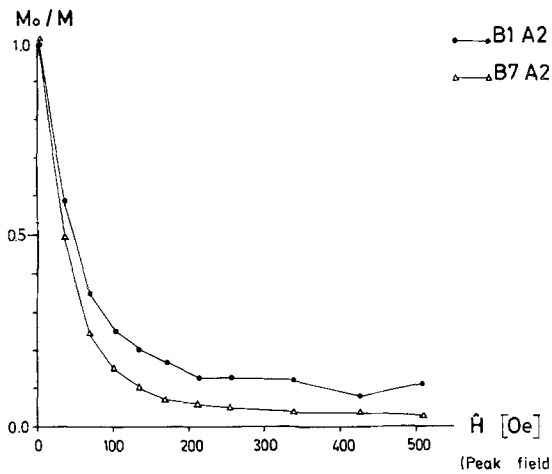


Figure 5
Alternating-field demagnetization-intensity curves. M/M_0 is the ratio of remaining to initial intensity after treatment in the corresponding peak alternating field. B1A2 is a specimen from site 1 and B7A2 a specimen from site 2

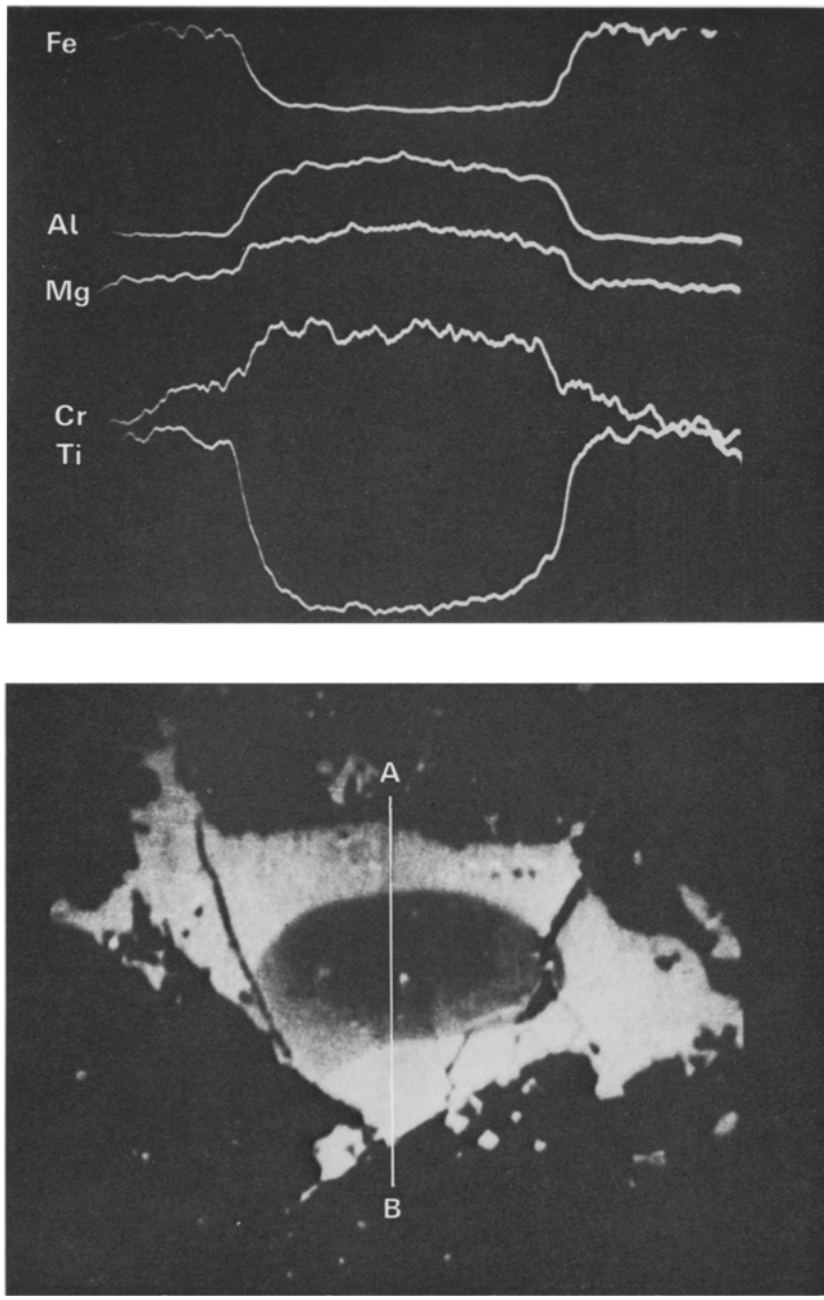


Figure 6
Electron microprobe analysis. Example of a titanomagnetite with a spinel core. Electron image of a crystal from specimen B10A with line scanning profiles of Fe, Ti, Al, Mg and Cr. Note the decrease of Fe, Ti and increase of the other elements in the darker part

elements such as Mg and Al lower the Curie point has been made using the data of FROHLICH *et al.* [15].) These predicted values were generally much lower than the measured ones. The method of analysis we adopted has been described by CREER and IBBETSON [8].

The electron beam was traversed across one of the grains having a grey core. The profile (Fig. 6) shows that the internal part is rich in Cr and Mn, and poor Ti. The outer part has approximately the same composition as the other titanomagnetites. BABKINE [4] has observed the same kind of phenomenon; the centre is attributed to a picotite, a spinel of the group [(Mg, Fe) (Al, $Cr)_2O_4$], (WINCHEL [21]). They suggest that the larger grains, poorer in Fe and Ti, and richer in trace elements could indicate an early formation or transformation of another spinel, e.g. a chromiferous spinel can form the nucleus of a titanomagnetite rich in Cr. This can be considered as an interesting explanation for our observations.

7. Thermomagnetic measurements

Thermomagnetic curves were obtained using a translation blance. The measurements above room temperature were achieved with an automatic continuous recording system. In the case of measurements from below room temperature to the Curie point, a manual compensating system with digital reading was used.

The thermomagnetic curves (Fig. 7) show that our specimens undergo irreversible changes on heating. The cooling curves differ from the initial heating curves in that there is an increase of the Curie temperature and the saturation magnetization J_s. The ratio of the saturation magnetization before and after heat treatment has a maximum value of two. OZIMA *et al.* [17] report a large increase of J_s after heating for submarine basalts and say that for continental basalts the known ratio seldom exceeds 50% of the original value. We think, however, that our results which show a large increase in J_s are typical of continental basalts containing homogeneous (i.e. unexsolved) titanomagnetites with low Curie points.

The Curie temperatures determined for the original and the heated specimens (up to 650°C) are listed in Table 6. All, except one, have a low Curie point indicating a large proportion of Ti in the titanomagnetites.

More detailed thermomagnetic measurements were made on one specimen from each site, by extending the analysis below room temperature. The curve (Fig. 8) shows an increase of the saturation magnetization from −180°C to approximately −50°C, where the value is maximum. Then a rapid decrease gives a Curie point at 200°C and, before the saturation value vanishes to zero, a small increase is marked at 400°C. OZIMA *et al.* [17], CREER and PETERSEN [9] have observed this behaviour. The general shape is like a *P* type curve as classified by NÉEL [16] but we conclude that this behaviour is not true *P* type because the magnetization measured below zero degree is not the saturation one as measurements in higher fields (10 K Oe) show. This similarity has also been discussed by READMAN *et al.* [18], who also conclude that titanomagnetites with high titanium content cannot be considered as true *P* types.

Table 6

Bulgarian basalts, oxidation stage and Curie point

Site	Sample No.		Microscope obervations under reflected light. Max. magnification 600				Curie Point, °C		(J_s) heated
			Shape of the grains	Grain Size Max. (micron)	Oxidation Stage	Js(T) curve Type	Heating process	Cooling process	(J_s) original
1	B1	With some large skeletal grains	Subhedral	80	1	I	230	515	1.5
1	B2		Subhedral	100	1	I	240	505	1.4
1	B3		Subhedral	200	1	II	420	505	1.1
2	B4		Subhedral	120	1	I	280	530	2.0
2	B6		Subhedral	400	1	I	250	540	2.0
2	B7		Subhedral	160	1	I	280	490	1.8
3	B8		Subhedral	250	1	I	290	510	1.4
3	B9		Subhedral	400	1	I	250	505	1.7
3	B10		Subhedral	150	1	I	210	510	1.6

Oxidation stage (WILSON and WATKINS [20]
$J_s(T)$ curve (CREER and VALENCIO [7])

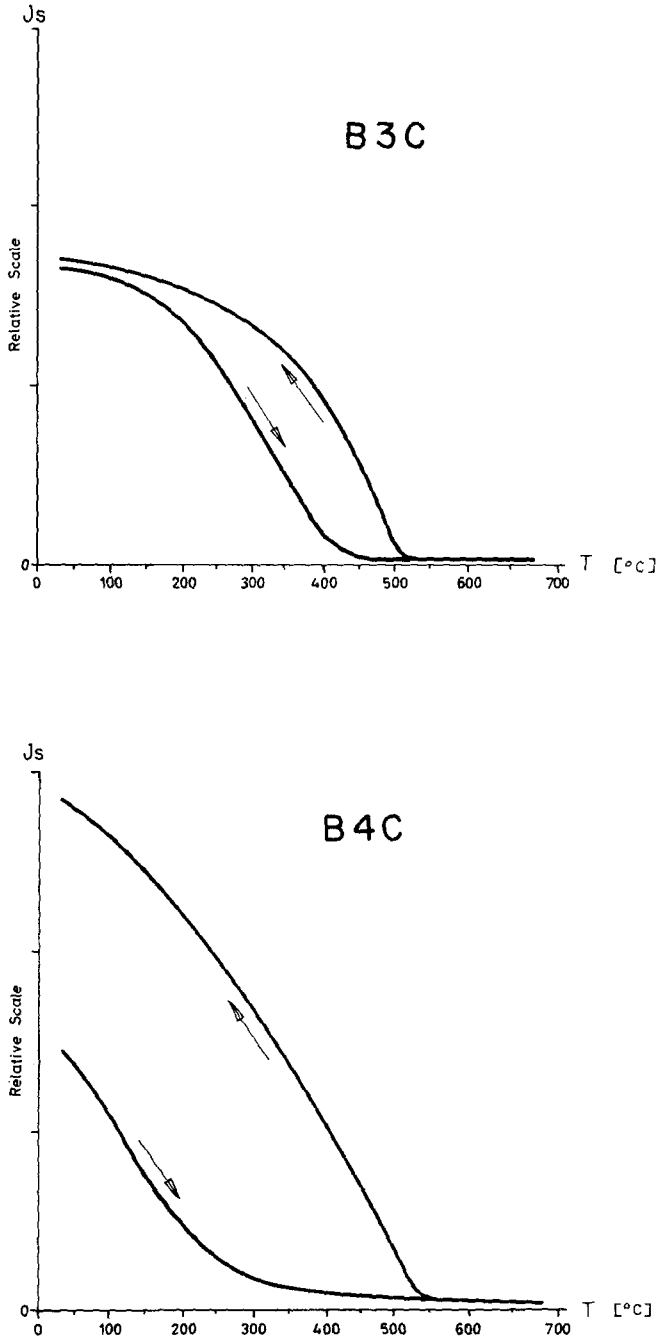


Figure 7
Some typical examples of thermomagnetic curves during heating and cooling. Magnetization is measured in arbitrary units

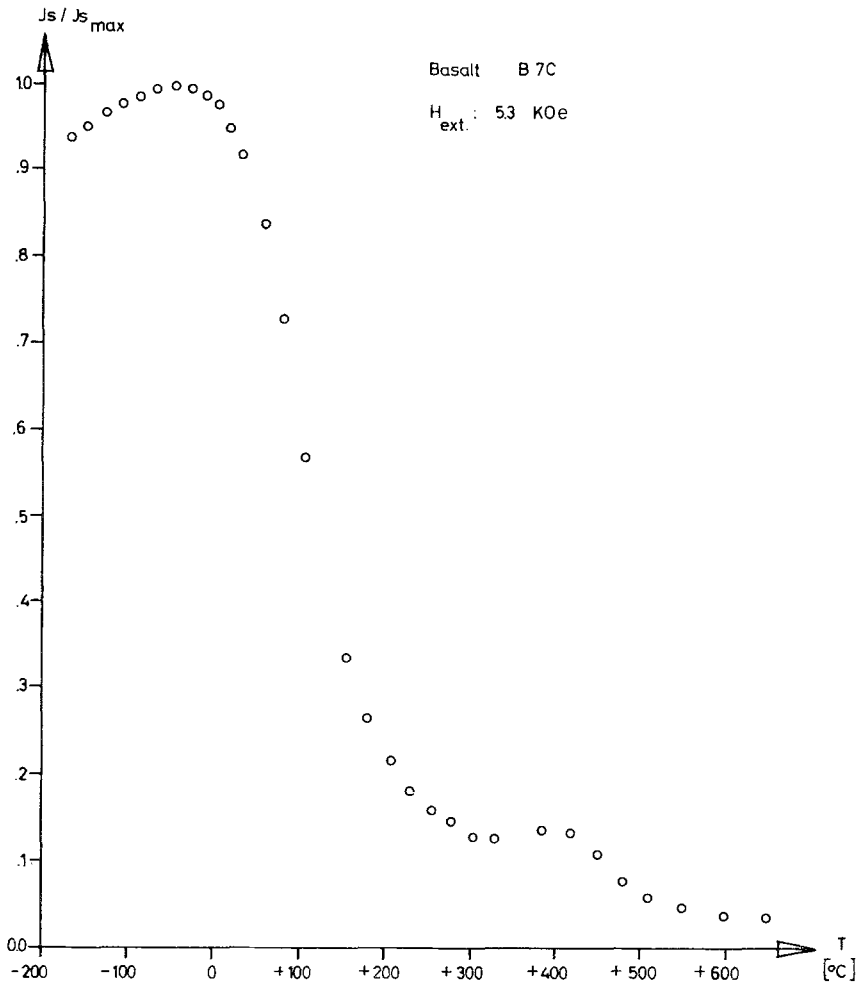


Figure 8
Thermomagnetic curve during heating from below room temperature measured on specimen B7C at $H = 5.3$ K Oe. Note the growth of magnetization due to the development of a new magnetic phase at 400°C

This apparent P type behaviour was found by CREER and PETERSEN [9] to progressively disappear after successively longer heat treatments in air at 400°C. This property might prove a useful indication of the extent of oxidation of natural 'titanomagnetites' in basalts without having to heat them.

8. *Discussion*

Our experiments on normal and reversely magnetized samples of these basalts lead us to the following conclusions:

(a) The 'titanomagnetites' appear homogeneous when observed under the ore microscope
(b) There is a high content of Ti.
(c) The Curie point and the saturation magnetization are increased by heating in air.
(d) The thermomagnetic curves measured below ambiant temperatures are apparently *P* type.
(e) A number of titanomagnetite grains possess spinel cores with a high Cr content.

The 'compositional' Curie points are lower than the measured ones. This cannot be explained by the presence of trace elements which usually tend to depress the Curie point of a pure titanomagnetite. We must stress, however, that the 'compositional' Curie points were deduced assuming stoichiometric titanomagnetite from the series $TiFe_2O_4$ (Ulvöspinel) – Fe_3O_4 (Magnetite). The observation that the measured Curie points are higher than expected on this assumption may be understood when we refer to the Curie temperature and the chemical composition diagram measured for cation deficient spinels (AKIMOTO *et al.* [3]). One sees that the composition of our samples could be represented by points situated in the area between the magnetite-ulvöspinel join and the Fe_2O_3–$FeTiO_3$ join of the TiO_2, FeO, compositional triangle. Our samples must then be oxidized in their natural state from a cubic spinel phase to a cubic cation deficient phase and we conclude that optical examination does not give a complete idea of their oxidation state. It has been observed that some trace elements e.g. Al, Mg may delay exsolution (FREUND *et al.* [14]), and the presence of such trace elements in our basalts may possibly account for our conclusion that the 'titanomagnetites' contained have become oxidized, in their natural state, to a homogeneous cation deficient phase rather than to composite grains containing exsolved patches of maghemite.

9. *Acknowledgements*

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